

VALLALAR ARTS AND SCIENCE COLLEGE.VADALUR

Department of Computer Science

A Project Report on

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM (SECMS)

Submitted by

Devadhatshini.B

Rani.S

Nishanthi.R

Geetha.V

B.SC(Computer Science)

Final Year

1. INTRODUCTION

1.1 Project Overview

The Student Enrollment & Course Management System (SECMS) is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. It enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting, all within a unified Salesforce environment. The project centralizes student, course, and enrollment data to ensure easy access, structured workflows, and accurate reporting, while enabling automated instructor assignment, task creation, validation checks, and multi-level approval processes.

1.2 Purpose

- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring

User Story

As an Enrollment Officer at the institute, I want to efficiently manage student admissions, course registrations, and instructor assignments using a centralized Salesforce application. I want the system to validate enrollment data automatically, route requests through an approval process before confirmation, and notify students of their enrollment status. I also want role-based dashboards and reports so that administrators and managers can monitor enrollment volume, revenue, and pending approvals in real time, enabling faster and more accurate decision-making.

2. IDEATION PHASE

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	Manage student enrollments efficiently	Enrollment data is scattered	Systems are manual and unintegrated	Frustrated and error-prone
PS-2	Administrator	Track enrollments and revenue	Reporting is delayed	Data is not centralized	Uncertain in decision-making

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

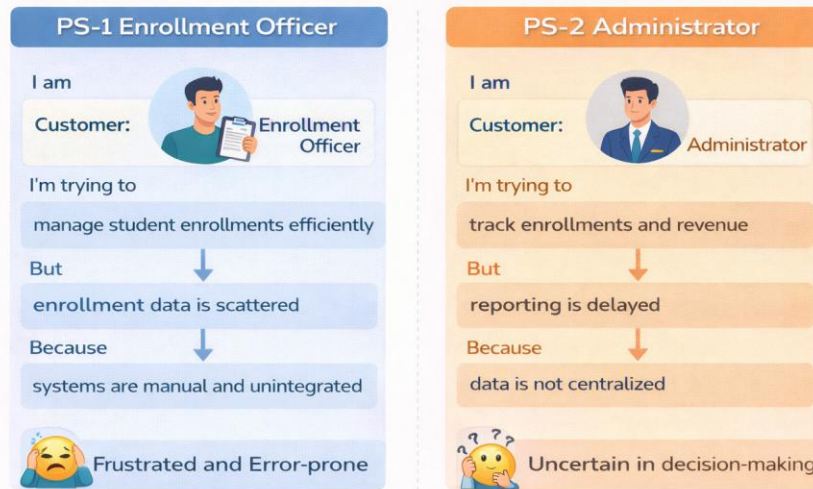


Fig 1: Problem Statements — SECMS

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

Aspect	Description
Think & Feel	Concerned about double bookings and worried about revenue loss
Hear	Complaints from students; feedback from management
See	Multiple spreadsheets and overflowing emails
Say & Do	Rechecks data manually and follows up by hand
Pain	Duplicate entries and approval bottlenecks
Gain	Automation, dashboards, and faster processing

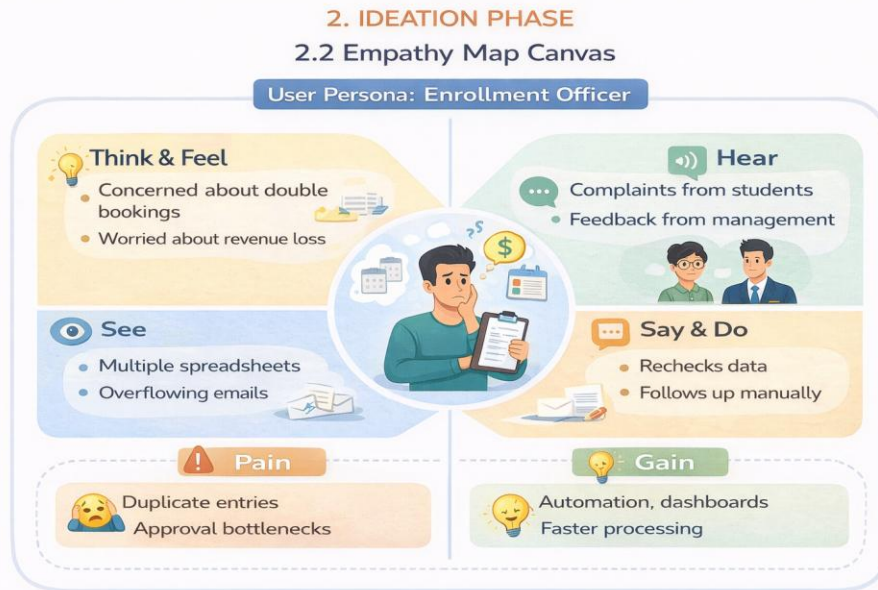


Fig 2: Empathy Map Canvas — SECMS

2.3 Brainstorming & Idea Prioritization

Brainstormed Ideas

- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Grouping

- Manual solutions — Spreadsheet automation
- Semi-automated solutions — Standalone enrollment portal, ERP integration
- Fully automated cloud solutions — Salesforce-based enrollment system

Selected Idea

Salesforce-based SECMS was selected due to high feasibility and high impact on solving enrollment officer and administrator pain points.

2.3 Brainstorming & Idea Prioritization — Student Enrollment & Course Management System (SECMS)

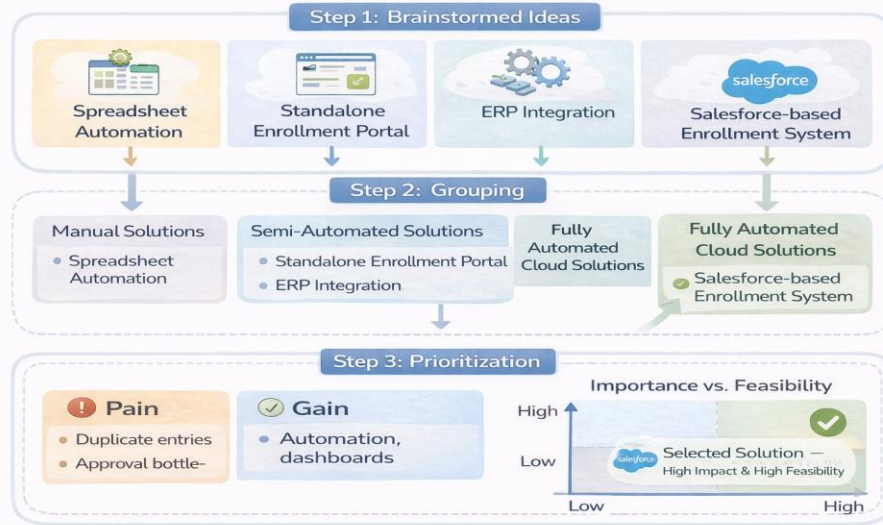


Fig 3: Brainstorming & Idea Prioritization — SECMS

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion

3.1 Customer Journey Map – Student Enrollment & Course Management System (SECMS)

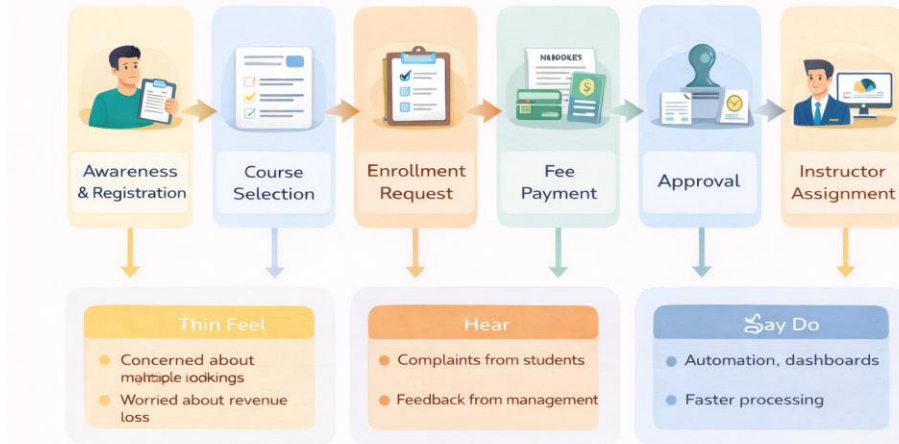


Fig 4: Customer Journey Map — SECMS

3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system provides a simple, intuitive, and user-friendly interface so that enrollment officers, administrators, instructors, and students can perform operations efficiently with minimal training, through clearly labeled tabs, dashboards, and forms — e.g. quick student registration, one-click enrollment approval, and role-based dashboards.

NFR-2: Security

The system ensures confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls — e.g. only admins can modify course details, instructors view only assigned students, and students cannot access other students' records.

NFR-3: Reliability

The system operates consistently and accurately during enrollment processing, approvals, and reporting, with automation logic that avoids duplicate or incorrect enrollments and produces consistent reports.

NFR-4: Performance

The system provides fast response times for all operations, including quick dashboard refresh, fast search for student records, and no lag during bulk uploads, especially during peak admission periods.

NFR-5: Scalability

The system supports future growth in students, courses, departments, and users without performance loss, allowing new departments, increased annual admissions, and extension to mobile or portal integration.

NFR-6: Availability

The system remains continuously accessible during institutional working hours with minimal downtime, offering 24/7 cloud access, reliability during peak admission season, and automatic backup of student records.

3.3 Data Flow Diagram

User inputs student and enrollment data → system validates and stores records → approval process routes the data → reports and dashboards consume the processed data. The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

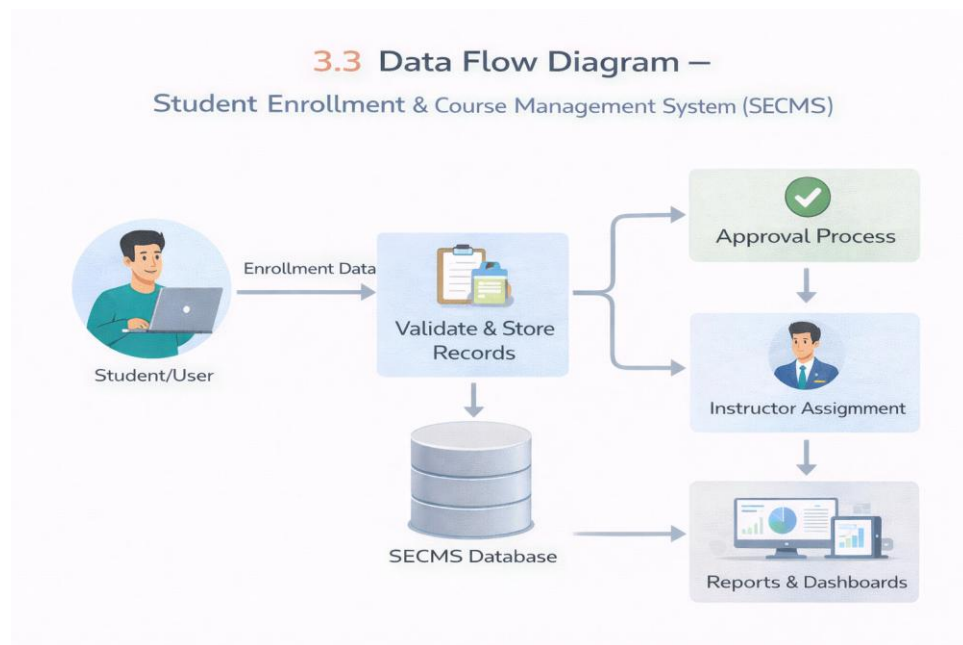


Fig 5: Data Flow Diagram — SECMS

3.4 Technology Stack

Layer	Technology
-------	------------

UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows — fragmented spreadsheets, manual paperwork, delayed approvals, and scattered systems — by aligning user behavior, constraints, and automation using Salesforce-native capabilities such as unified dashboards, automated approvals, and a centralized database.

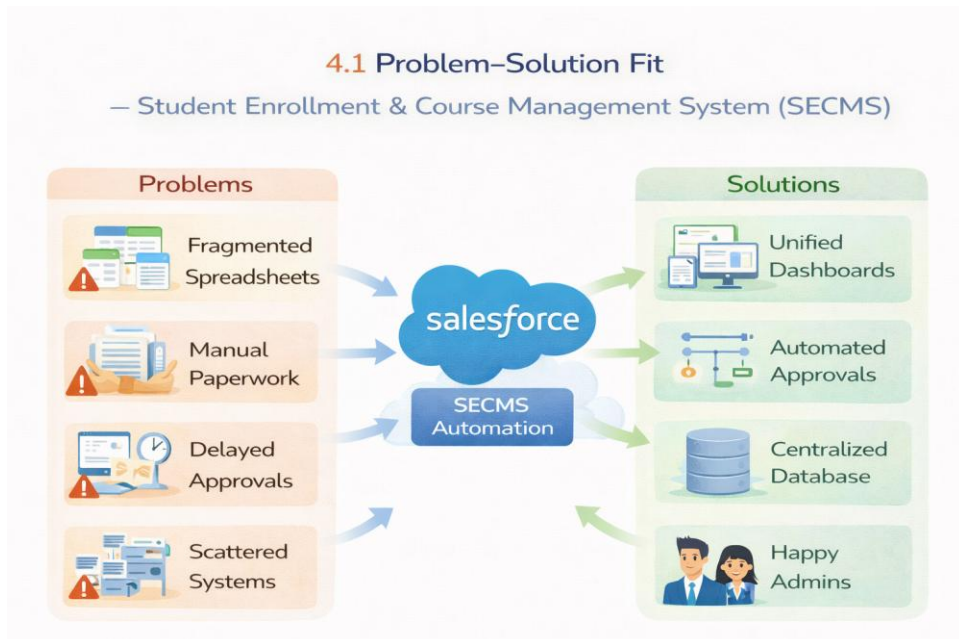


Fig 6: Problem–Solution Fit — SECMS

4.2 Proposed Solution

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience

Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates four Salesforce objects — Student, Course, Enrollment, and Instructor — with Automation Flows, Approval Processes, Dashboards, and a Security layer sitting on top of the Salesforce Platform.

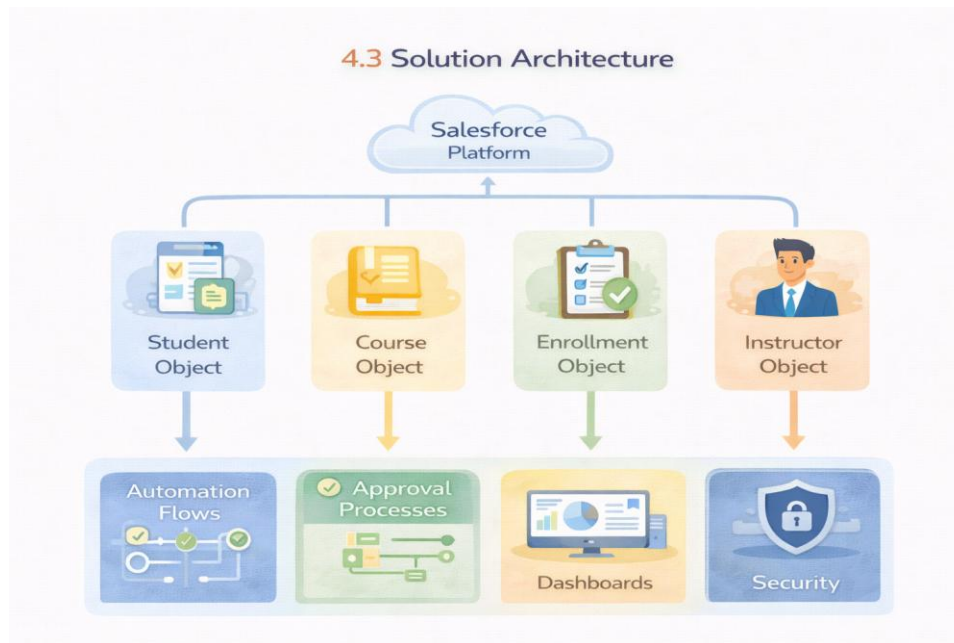


Fig 7: Solution Architecture — SECMS

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog & Sprint Schedule

Sprint	Epic	USN	User Story / Task	Points	Priority
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the system.	1	High
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and	3	High

			Instructor so that academic data can be stored and managed efficiently.		
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields so that complete records can be captured accurately.	5	High
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed.	5	High
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect records are prevented.	3	High
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced.	3	High
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can quickly analyze system performance.	4	Medium

Project Tracker, Velocity & Burndown

Sprint	Total Points	Duration	Start Date	Planned End Date
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022

6. PROJECT DEVELOPMENT PHASE

Project flow: Salesforce Developer Account Creation → Custom Object Creation → Tabs Creation → Lightning App Creation → Fields Creation → Relationships → Page Layout Customization → Validation Rules → Flows (Automation) → Approval Process → Reports → Dashboards → Security Setup.

Milestone 1 — Salesforce Developer Account

A free Salesforce Developer Edition org was created at developer.salesforce.com/signup using the institute's details, activated via the verification email, and secured with a password and security question, redirecting to the Salesforce Setup home page.

Fig 8: Sign up for Salesforce Developer Edition

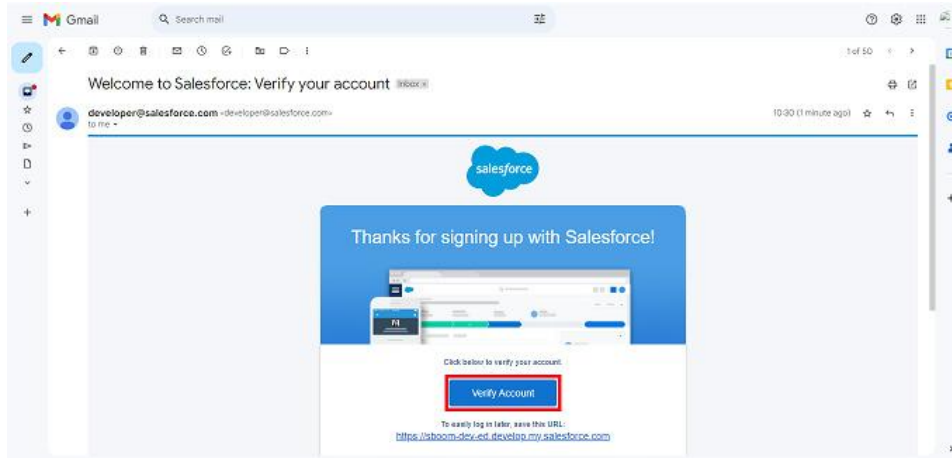


Fig 9: Verify Account email

Change Your Password

Enter a new password for lead@sb.com.
 Make sure to include at least:

- 8 characters
- 1 letter
- 1 number

* New Password
 Good

* Confirm New Password
 Match

Security Question

* Answer

Change Password

Fig 10: Change Your Password

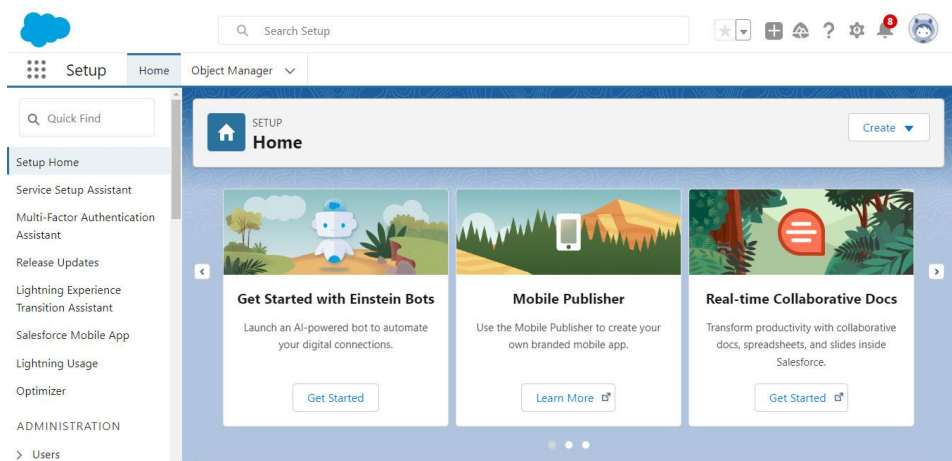


Fig 11: Salesforce Setup home page

Milestone 2 — Objects

Four custom objects — Student, Course, Instructor, and Enrollment — were created via Setup → Object Manager → Create → Custom Object, with Label, Plural Label, and a Text-type Record Name set for each (e.g. Student / Students / Student Name). Allow Reports, Allow Activities, Track Field History, and Allow Search were enabled for every object, and each was set to Deployed status.

The screenshot shows the 'Custom Object Definition Edit' page for the 'Student' object. The left sidebar contains a navigation menu with options like 'Details', 'Fields & Relationships', 'Page Layouts', etc. The main content area is titled 'Custom Object Information' and includes fields for 'Label' (Student), 'Plural Label' (Students), 'Object Name' (Student), and 'Description'. A warning message states: 'The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.' Below this, there are checkboxes for 'Starts with vowel sound' and 'Context-Sensitive Help Setting'. At the bottom, there are radio buttons for 'Open the standard Salesforce.com Help & Training window' and 'Open a window using a Visualforce page'.

Fig 12: Custom Object Definition Edit — Student

The screenshot shows the 'Enter Record Name Label and Format' page for the 'Student' object. The left sidebar is the same as in Fig 12. The main content area has a section for 'Record Name' with a text input field containing 'Student Name' and a dropdown for 'Data Type' set to 'Text'. Below this is the 'Optional Features' section with checkboxes for 'Allow Reports', 'Allow Activities', 'Track Field History', 'Allow in Chatter Groups', and 'Enable Licensing'. The 'Object Classification' section has checkboxes for 'Allow Sharing', 'Allow Bulk API Access', and 'Allow Streaming API Access'. The 'Deployment Status' section has radio buttons for 'In Development' and 'Deployed' (selected). A warning message states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type'.

Fig 13: Record Name Label and Format — Student

The screenshot shows the 'Deployment Status' and 'Search Status' section for the 'Student' object. The 'Deployment Status' section has radio buttons for 'In Development' and 'Deployed' (selected). The 'Search Status' section has a checkbox for 'Allow Search' which is checked. At the bottom, there are buttons for 'Save', 'Save & New', and 'Cancel'.

Fig 14: Deployment Status and Search Status — Student

Milestone 3 — Fields & Relationships

Student Object fields

Student Name (Text, Standard); Email (Email); Phone (Phone); Date of Birth (Date); Student Status (Picklist); Address (Text Area).

Course Object fields

Course Name (Text, Standard); Category (Picklist); Duration (Number/Text); Course Fee (Currency); Description (Text Area).

Instructor Object fields

Instructor Name (Text, Standard); Instructor Code (Text); Expertise (Picklist); Phone (Phone); Email (Email).

Enrollment Object fields

Student (Lookup — Student); Course (Lookup — Course); Instructor (Lookup — Instructor); Enrollment Date (Date); Enrollment Status (Picklist); Fees Paid (Checkbox); Total Amount (Currency); Comments (Text Area).

Relationships

Three lookup relationships were created on the Enrollment object: Enrollment → Student, Enrollment → Course, and Enrollment → Instructor, each added to the respective page layouts with a child relationship name of "Enrollments" so related enrollment records are visible from Student, Course, and Instructor records.

The screenshot shows the 'New Enrollment Custom Field' setup for 'Student'. The 'Data Type' is set to 'Lookup Relationship'. The 'Next' button is highlighted.

Fig 15: New Lookup Relationship — choosing the data type

The screenshot shows the 'Enrollment → Student lookup field' setup. The 'Field Label' is 'Student', the 'Child Relationship Name' is 'Enrollments', and the 'Save' button is highlighted.

Fig 16: Enrollment → Student lookup field

The screenshot shows the 'New Enrollment Custom Field' setup for 'Course'. The 'Data Type' is set to 'Lookup Relationship'. The 'Next' button is highlighted.

Fig 17: New Relationship — choosing the related object (Course)

Enrollment

Step 2. Enter the label and name for the lookup field Step 2 of 4

Field Label

Field Name

Description

Help Text

Child Relationship Name

Required ☒ Always require a value in this field in order to save a record

What to do if the lookup record is deleted? ☐ Clear the value of this field. You can't choose this option if you make this field required.

Previous Save Next Cancel

Fig 18: Enrollment → Course lookup field

Enrollment

Fields & Relationships

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected Select one of the data types below.

☐ Auto Number A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☐ Formula A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☒ Lookup Relationship Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Next Cancel

Fig 19: New Relationship — Fields & Relationships panel

Enrollment

New Relationship

Step 2. Choose the related object Step 2

Select the other object to which this object is related.

Related To

Previous Next Cancel

Fig 20: Choosing the related object (Instructor)

Enrollment

New Relationship

Step 3. Enter the label and name for the lookup field Step 3 of 6

Field Label

Field Name

Description

Help Text

Child Relationship Name

Required ☒ Always require a value in this field in order to save a record

What to do if the lookup record is deleted? ☐ Clear the value of this field. You can't choose this option if you make this field required.

☒ Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity.

Previous Next Cancel

Fig 21: Enrollment → Instructor lookup field

Enrollment

New Relationship

Step 6. Add custom related lists Step 6 of 6

Field Label Instructor

Data Type Lookup

Field Name Instructor

Description

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

Add Related List Page Layout Name

☒ Instructor Layout

☒ Append related list to users' existing personal customizations

Previous Save & New Save Cancel

Fig 22: Saving the Enrollment → Instructor relationship

Milestone 4 — Validation Rules

Object	Rule Name	Purpose
Student	Student_Email_Domain_Validation	Ensures students use only a valid institutional email address (e.g. @student.edu)
Enrollment	Fees_Must_Be_Paid_Before_Approval	Prevents an enrollment from being approved until fees are marked as paid

SETUP > OBJECT MANAGER Student

Validation Rule Edit

Rule Name: Email_Must_Student_Domain

Active: ☒

Description:

Error Condition Formula

Example: Discount_Percent_c > 0.30
Display an error if Discount is more than 30%
If this formula expression is true, display the text defined in the Error Message area

Insert Field Insert Operator

NOT (CONTAINS (Email_c, "@student.edu"))

Check Syntax

Functions

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function

ABS(number)
Returns the absolute value of a number, a number without its sign
Help on this function

Save Save & New Cancel

Fig 23: Student Email Domain validation rule

Error Message

Example: Discount percent cannot exceed 30%

This message will appear when Error Condition formula is true

Error Message: Email must be a valid student institution email

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☐ Top of Page ☒ Field Email

Save Save & New Cancel

Fig 24: Validation rule error message — Email

Setup > OBJECT MANAGER

Enrollment

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Validation Rules

Rule Name: **Fees_Must_Be_Paid_Before_Approval**

Active: ☒

Description:

Error Condition Formula

Example: `Discount_Percent__c > 0.30` More Examples...
 Display an error if Discount is more than 30%.
 If this formula expression is **true**, display the text defined in the Error Message area

Insert Field: `AND ( ISPIKEDVAL (Enrollment_Status__c, "Approved"), Fee_Paid__c = FALSE )`

Insert Operator:

Check Syntax

Functions:
 -- All Function Categories --
 ABS
 ACOS
 ADDMONTHS
 AND
 ASCII
 ASIN
 Insert Selected Function
 ABS(number)
 Returns the absolute value of a number, a number without its sign
[help on this function](#)

Fig 25: Fees Must Be Paid Before Approval — formula

Error Message

Example: `Discount percent cannot exceed 30%`

This message will appear when Error Condition formula is **true**

Error Message: `Fees must be paid before approving an enrollment`

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☒ Top of Page ☐ Field

Save Save & New Cancel

Fig 26: Validation rule error message — Fees

Milestone 5 — Approval Process

An approval process (Approval_Enrollment_Request) was configured on the Enrollment object with entry criteria Enrollment_Status__c = "Requested", routing requests to the Training Manager for approval. On final approval, a field update sets Enrollment_Status__c to "Approved"; on rejection, a field update sets it to "Rejected" and an email alert notifies the student. The process was then activated.

Setup > Home > Object Manager > Enrollment

Approval Processes

Open Approvals App

Approvals are complex business processes that require information gathering and planning before implementing. It is recommended that you follow the instructions below before getting started.

1. Read the hello topic
2. View the checklist
3. Create a custom user hierarchical relationship field
4. Create email templates
5. Create an approval process using either the Jump Start or Standard Wizard
6. Add Approval History Related List to all page layouts
7. Activate the process to deploy to your users

Manage Approval Processes For: **Enrollment**

Create New Approval Process

Use Jump Start Wizard

Use Standard Setup Wizard

Fig 27: Approval Processes setup — Enrollment

Approval Process Edit
Approval Enrollment Request

Step 1. Enter Name and Description

Enter a name and description for your new approval process.

Enter Name and Description

Process Name	Approval Enrollment Request
Unique Name	Approval Enrollment Request
Description	Approval process for Enrollment requests — sends to Training Manager

Fig 28: New Approval Process — name and description

Step 2. Specify Entry Criteria

Previous Save **Next**

If only certain types of records should enter this approval process, enter that criteria below. For example, only expense reports from employees at headquarters should use this approval process.

Specify Entry Criteria

Use this approval process if the following criteria are met

Field	Operator	Value	
Enrollment: Enrollment Status	equals	Requested	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

Fig 29: Specify Entry Criteria

SETUP
Approval Processes

Approval Step Edit
Training Manager

Help for this Page

Step 1. Enter Name and Description Step 1 of 3

Save Next Cancel

Enter a name, description, and step number for your new approval step.

Enter Name and Description Required Information

Approval process Name: Approval enrollment request

Name	Training Manager
Unique Name	Training_Manager
Description	

Save Next Cancel

Fig 30: Approval step — Training Manager


Field Updates

Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the fields shown only for the type that you select.

Field Update Edit
Save Save & New Cancel

Identification

Name

Unique Name

Description

Object Enrollment
Field to Update Enrollment: Enrollment Status
Field Data Type Picklist
Re-evaluate Workflow Rules after Field Change ☐

Specify New Field Value

Picklist Options

☐ The value above the current one
☐ The value below the current one
☒ A specific value

Save Save & New Cancel

Fig 31: Field Update — Status to Approved

Field Update Edit
Save Save & New Cancel

Identification

Name

Unique Name

Description

Object Enrollment
Field to Update Enrollment: Enrollment Status
Field Data Type Picklist
Re-evaluate Workflow Rules after Field Change ☐

Specify New Field Value

Picklist Options

☐ The value above the current one
☐ The value below the current one
☒ A specific value

Fig 32: Field Update — Set Status to Rejected

Email Alert Edit
Save Save & New Cancel

Edit Email Alert

Description

Unique Name

Object Enrollment

Email Template

Protected Component ☐

Recipient Type Search: for: Find

Available Recipients

User: Integration User ▲
User: OrgFarm EPIC
User: Security User

Add
Remove

Selected Recipients

User: sai kumar A ▲

Fig 33: Email Alert Edit — Enrollment Rejected

Fig 34: Email Alert — recipients and From address

Milestone 6 — Tabs

Custom object tabs were created for Student, Course, Instructor, and Enrollment via Setup → Tabs → New Custom Object Tab, each with a chosen tab style and made visible to all profiles.

Fig 35: Custom Object Tab creation — Student

Milestone 7 — The Lightning App

A Lightning App named "Student Enrollment & Course Management" was created via App Manager → New Lightning App, assigned to the Admin, Enrollment Officer, and Instructor profiles, with Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, and Tasks added as navigation items.

Fig 36: Lightning Experience App Manager — New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

*App Name ⓘ

*Developer Name ⓘ

Description ⓘ

App Branding

Image ⓘ



Primary Color Hex Value ⓘ

Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

Fig 37: App Details & Branding

Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

Available Items

🔍 Type to filter list...

- Action Hub
- Activation Targets
- Activations
- Agentforce Studio
- All Sites
- Alternative Payment Methods
- Analytics

Selected Items

- Students
- Courses
- Instructors
- Enrollments
- Accounts
- Contacts
- Dashboards

Fig 38: Navigation Items — selecting custom objects

User Profiles

Choose the user profiles that can access this app.

Available Profiles

🔍 system Administration

Selected Profiles

- System Administrator
- Standard User

Fig 39: User Profiles — assigning the app

Milestone 8 — Page Layouts and Record Types

Two record types — New Enrollment and Re-Enrollment — were created on the Enrollment object to separate first-time and repeat enrollment processes, each enabled for the Admin and Enrollment Officer profiles. Matching page layouts (New Enrollment Layout and Re-Enrollment Layout) were

cloned from the base Enrollment layout, and Enrollment-related lists were added to the Student, Course, and Instructor page layouts so related enrollments are visible from each parent record.

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Edit Record Type
Re-Enrollment

Enter a new name for the selected record type and click Save.

Record Type

Record Type Label: Re-Enrollment

Record Type Name: Re_Enrollment

Namespace Prefix:

Description: Used when a student re-enrolls in a course

Active: ☒

Save Cancel

Fig 40: Record Type — Re-Enrollment

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Edit Record Type
New Enrollment

Enter a new name for the selected record type and click Save.

Record Type

Record Type Label: New Enrollment

Record Type Name: New_Enrollment

Namespace Prefix:

Description: Used for first-time enrollments

Active: ☒

Save Cancel

Fig 41: Record Type — New Enrollment

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Create New Page Layout

As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard fields and relationships.

Existing Page Layout: Enrollment Layout

Page Layout Name: New Enrollment Layout

Save Cancel

Fig 42: Create New Page Layout — New Enrollment

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Quick Find: Field Name

Fields

Buttons

Quick Actions

Mobile & Lightning

Actions

Expanded Lookups

Related Lists

Report Charts

Section

Blank Space

Course Fee	Enrollment Status	Owner
Created By	Fees Paid	Record Type
Enrollment Name	Instructor	Student
Enrollment Date	Last Modified By	

Information (Header visible on edit only)

- Enrollment Name: Sample Text
- Student: Sample Text
- Course: Sample Text
- Instructor: Sample Text
- Enrollment Date: 12/22/2025
- Enrollment Status: Sample Text
- Fees Paid: ☒
- Comments: Sample Text
- Course Fee: \$123.45

Save

Fig 43: New Enrollment layout — fields arranged

Setup > Object Manager > Enrollment

Create New Page Layout

As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard sections whose

Existing Page Layout: Enrollment Layout

Page Layout Name: Re-Enrollment Layout

Save Cancel

Fig 44: Create New Page Layout — Re-Enrollment

Setup > Object Manager > Enrollment

Page Layouts

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Fields

Quick Find: Field Name

Section	Field Name	Value	Owner
Blank Space	Created By	Fees Paid	Record Type
	Enrollment Name	Instructor	Student
	Enrollment Date	Last Modified By	

Information (Header visible on edit only)

- Enrollment Name: Sample Text
- Student: Sample Text
- Course: Sample Text
- Enrollment Status: Sample Text
- Fees Paid: ✓
- Enrollment Date: 12/22/2025
- Comments: Sample Text
- Course Fee: \$123.45
- Instructor: Sample Text

Owner: Sample Text

Fig 45: Re-Enrollment layout — fields arranged

Milestone 9 — Automation with Flow

A record-triggered flow (SetEnrollmentDate) sets Enrollment_Date__c to the current date whenever an Enrollment record is created. A second flow sends an email alert to the student, notifies the admin, and sets the related Student's status to "Active" whenever Enrollment_Status__c changes to "Approved". A third flow (AutoAssign_Instructor) reads the related Course's category and automatically assigns Instructor A, B, or C for Technical, Language, and Non-Technical courses respectively. A fourth flow (Create_Followup_Task) creates a follow-up Task for the Admin, due two days out, whenever an enrollment is set to "Requested".

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Enrollment

Configure Trigger

Trigger the Flow When:

- ☒ A record is created
- ☐ A record is updated
- ☐ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of records that trigger the flow to conserve your org's resources.

Condition Requirements

None

Fig 46: Record-Triggered Flow — Set Enrollment Date

 Update Records

* Label

Set Enrollment Date

* API Name ⓘ

Set_Enrollment_Date

Description

*How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow

☐ Update records related to the enrollment record that triggered the flow

☐ Use the IDs and all field values from a record or record collection

☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record

Set Field Values for the Enrollment Record

Field

 Enrollment_Date X

←

Value

 Running Flow Interview > CurrentDate X



Fig 47: Update Records — Set Enrollment Date

Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

☐ A record is created
☐ A record is updated
☒ A record is created or updated
☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Approved

Fig 48: Flow trigger — Enrollment Status equals Approved

Get Records

* Label

Get Student

* API Name

Get_Student

Description

Get Records of This Object

*** Data Source**

☐ Data Cloud Object
☒ Salesforce Object

*** Object**

Student

Filter Student Records

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Status	Equals	Active

Fig 49: Get Records — related Student

 Enrollment Approved EmailAlert
 ✕

* Label

Send Student Email

* API Name ⓘ

Send_Student_Email

Description

Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

A_a * Record ID ⓘ

A_a Triggering Enrolement_c > Record ID ✕

Fig 50: Enrollment Approved Email Alert — Send Student Email

 Update Records
 ✕

* Label

Set Student Status Active

* API Name ⓘ

Set_Student_Status_Active

Description

* How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow
 ☐ Update records related to the enrollment record that triggered the flow
 ☐ Use the IDs and all field values from a record or record collection
 ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record ▼

Set Field Values for the Enrollment Record

Field	Value
A _a Comments ✕	←  True ✕

Fig 51: Update Records — Set Student Status Active

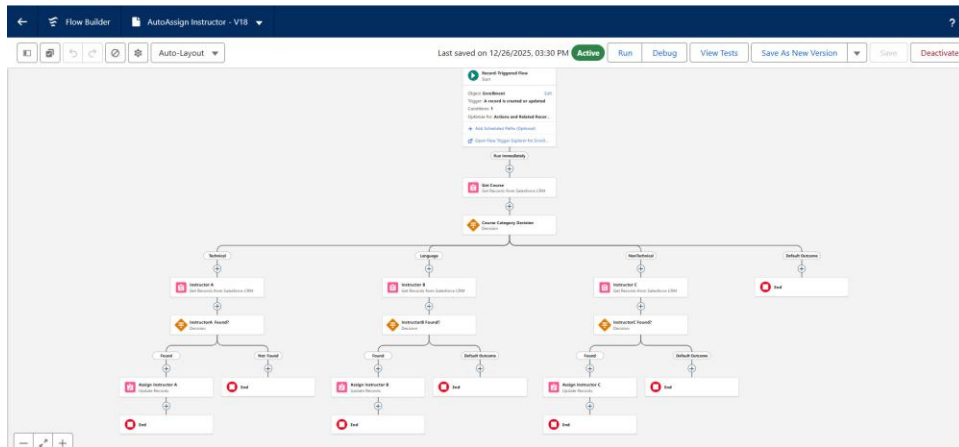


Fig 52: AutoAssign Instructor — full flow diagram

Configure Start

*Object

Enrollment

Configure Trigger

Trigger the Flow When:

☐ A record is created
 ☐ A record is updated
 ☒ A record is created or updated
 ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field

Enrollment Status

Operator

Equals

Value

A Requested

Fig 53: Create_Followup_Task — flow trigger

Get Records

Get Admin User | Get_Admin_User

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object

☒ Salesforce Object

* Object

User

Filter User Records

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
A Username	Equals	geespam198430@agentforce.com

+ Add Condition

Fig 54: Get Records — Get Admin User

Create Records

Create a Record of This Object

* Object

Task

Set Field Values for the Task

Field	Value
Due Date Only	DueDateTodayPlus2
Description	Auto-created follow-up for enrollment request
Priority	Normal
Status	Not Started
Subject	Follow up on enrollment request
Related To ID	Triggering Enrolment_c > Record ID

Fig 55: Create Records — follow-up Task

Milestone 10 — Reports

Four reports were built: Students by Status (Students report type, grouped by Status__c), Enrollments by Course (Enrollments report type, grouped by Course__r.Name), Pending Enrollment Approvals (Enrollments filtered to Enrollment_Status__c = "Requested"), and Revenue Report – Fees Paid (Enrollments filtered to Fees_Paid__c = TRUE, grouped by course, summarizing the course fee).

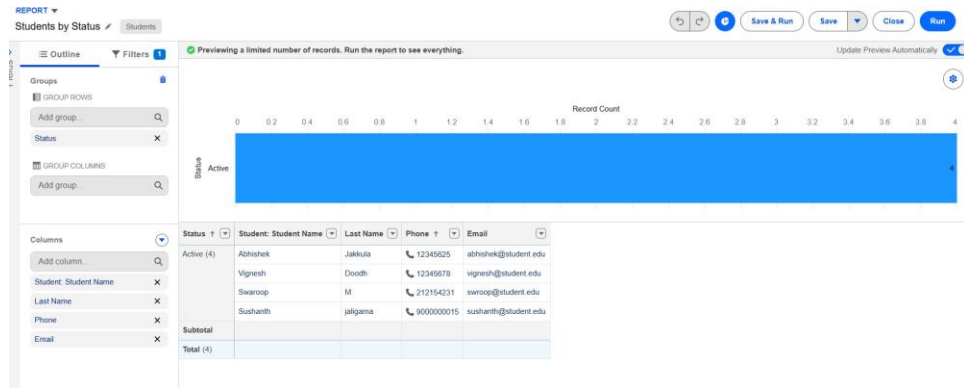


Fig 56: Students by Status report

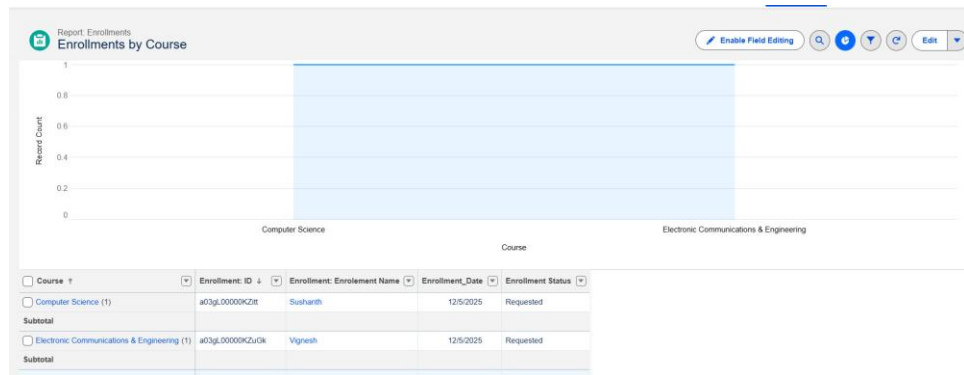


Fig 57: Enrollments by Course report

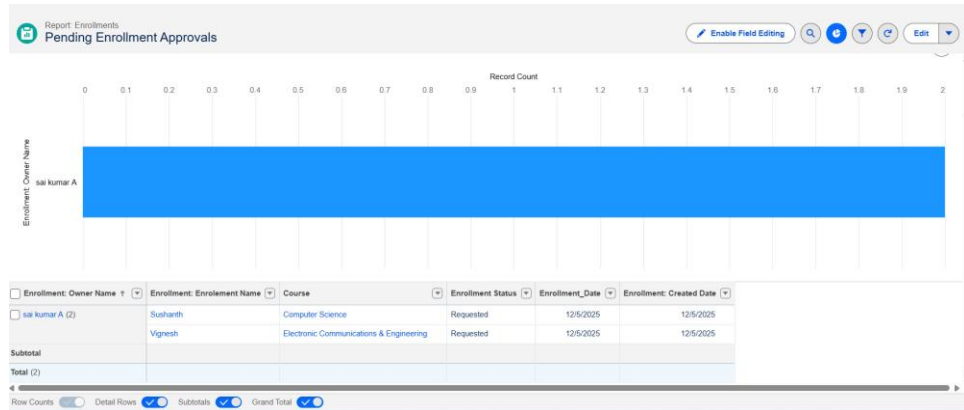


Fig 58: Pending Enrollment Approvals report

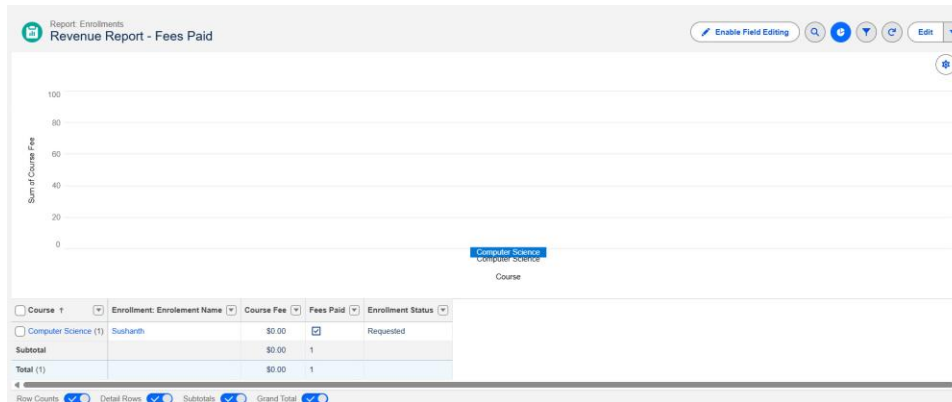


Fig 59: Revenue Report — Fees Paid

Milestone 11 — Dashboards

A "Student Management Dashboard" was created combining a Donut Chart (Students by Status), a Bar Chart (Enrollments by Course), a Gauge (Revenue Report – Fees Paid), and a Table (Pending Enrollment Approvals) on a single canvas for at-a-glance monitoring.

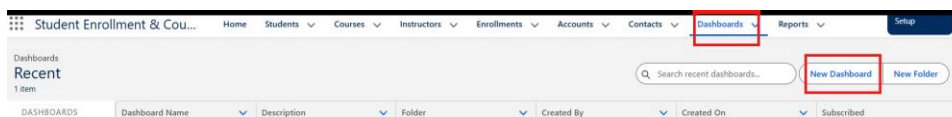


Fig 60: Dashboards navigation — Lightning App

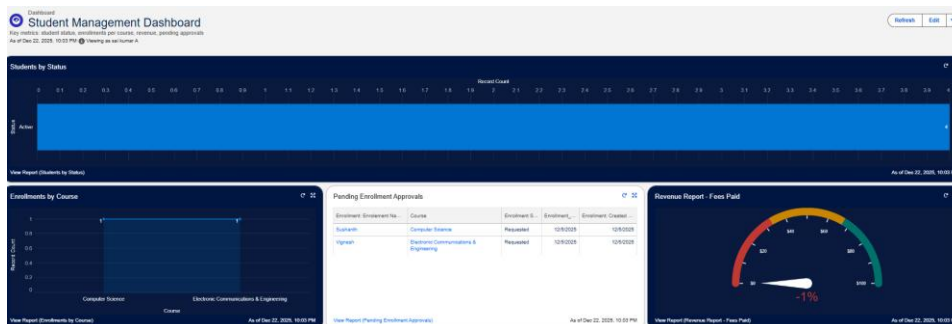


Fig 61: Student Management Dashboard

Milestone 12 — Security Setup

Profiles were configured for Admin (full access), Enrollment Officer (create/edit access to Enrollment, read/edit on Student and Course), and Instructor (read-only access to assigned Enrollments). Permission sets were created for additional object access and dashboard editing. Organization-wide defaults were set to Public Read/Write for Student and Course, and Private for Enrollment, with an Instructor lookup on Enrollment and a supporting flow used to restrict instructors to viewing only their own assigned records.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from the Salesforce interface to verify: student, course, instructor, and Enrollment record creation; validation rule checks for mandatory fields and Enrollment constraints; record type behavior for New Enrollment vs Re-Enrollment; flow execution

for automated instructor assignment and approval notifications; and generated reports and dashboards for accurate data visualization and performance monitoring. The configuration screenshots supporting each milestone are included inline in Section 6 above.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. CONCLUSION

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle — from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce's automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making, demonstrating how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. APPENDIX

Item	Description
Source Code	Salesforce Configuration
Dataset	Salesforce Objects
Tools Used	Salesforce Developer Org